

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

SSC PART-I (CLASS 9) PRACTICE EXAMINATION

CHEMISTRY PAPER 24 | ANSWER KEY (Version A)

Syllabus: Chapter 11 – Environmental Chemistry: Air (35 marks) + Revision: Chapters 8, 10 (15 marks)

Total Marks: 50

New chapter: 35

Revision: 15

Marking rule used on this paper: a missing **name**, a missing **unit**, or a missing set of **state symbols** where they were asked for, loses $\frac{1}{2}$ **mark**, the same in every question. It never cancels a correct equation or correct working.

SECTION A: ANSWER KEY ($10 \times 1 = 10$ MARKS)

1. Percentage of oxygen in clean, dry air: (b) 21% 1

Clean, dry air is about **78% nitrogen** and **21% oxygen**; the remainder is noble gases and carbon dioxide. 78% is the trap; that is nitrogen.

Source: Ch 11, section 11.1 and Fig. 11.1, book p151. SLO: composition of clean, dry air.

2. Besides acid rain, sulphur dioxide is responsible for: (a) haze 1

The book: "Sulphur dioxide also responsible for acid rain and haze." Melting of polar ice is an effect of **global warming**, caused by CO_2 and methane, not SO_2 .

Source: Ch 11, section 11.2.1, book p152. SLO: adverse effects of air pollutants, (f) sulphur dioxide: acid rain and haze.

3. Particulate matter is harmful mainly because it can: (c) penetrate deep into the lungs, causing respiratory problems and lung cancer 1

Particulates are fine particles such as soot, smoke and dust. Option (d) describes **carbon monoxide**, which also comes from incomplete combustion; keep the two apart.

Source: Ch 11, section 11.2.4, book p153. SLO: (c) particulates, increased risk of respiratory problems and cancer.

4. Nearly half of all air pollution comes from: (d) cars and other motor vehicles 1

The book: "Nearly half of the air pollution comes from cars and other motor vehicles." Volcanoes, forest fires, termites and cows are natural sources.

Source: Ch 11, section 11.3.2, book p154.

5. A catalytic converter contains: (b) platinum, palladium and rhodium 1

Calcium oxide is the trap: it is used in **flue gas desulphurization** in power plants, not in a car's exhaust.

Source: Ch 11, section 11.6, book p158. SLO: removal of oxides of nitrogen by catalytic converters.

6. An important source of methane: (a) rice fields 1

Wetlands, **rice fields**, landfills and livestock are sources of methane. Car engines and lightning produce **nitrogen oxides**; coal-burning power stations produce **sulphur dioxide**.

Source: Ch 11, section 11.2.5, book p153. SLO: (c) methane from the decomposition of vegetation and waste gases from digestion in animals.

7. Photosynthesis removes from the air: (c) carbon dioxide 1

Photosynthesis: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. That is why planting trees is a strategy against global warming.

Source: Ch 11, sections 11.7 and 11.8, book p158–160. SLO: define photosynthesis.

8. [REVISION, Ch 10] Correct ionization of nitric acid: (b) $\text{HNO}_3 \rightarrow \text{H}^+ + \text{NO}_3^-$ 1

Ionization means the acid **breaks up** into ions, so the acid is on the left. Option (a) is written backwards, which is exactly the slip in exam-23 Q3(i). Strong acid, so a single arrow.

Re-test of exam-23 Q3(i) and (ii), both written in reverse. Ch 10 SLO: formulate dissociation equations.

9. [REVISION, Ch 10] Bronsted-Lowry acid in $\text{H}_2\text{S} + \text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{HS}^-$: (d) H_2S 1

H_2S **donates** a proton and becomes HS^- (a negative ion). NH_4^+ is the trap: it is a **product**, formed when NH_3 accepts the proton. Choose the acid from the left side of the arrow.

Source: Ch 10, Concept Assessment 10.1 Q3, book p144. Re-test of exam-23 Q2(ii), where the product H_3O^+ was named as the acid.

10. [REVISION, Ch 10] Magnesium burning in oxygen: (a) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ 1

Oxygen gas is always O_2 . Option (b) balances only because it wrongly uses a single O atom.

Source: Ch 10, section 10.5, book p146. Re-test of exam-23 Q8(i), " $2\text{Mg} + \text{O} \rightarrow 2\text{MgO}$ ".

SECTION B: ANSWER KEY ($6 \times 4 = 24$ MARKS)

2. Pollutants and their sources. 4

(i) "Anything that is in the air, water or soil which has a **harmful effect** on some part of the environment is called a pollutant." 1

(ii) Any **two**, 1 mark each (source **and** its pollutant):

- Forest fires and dust storms → smoke and dust particles
- Volcanoes → clouds of dust, poisonous gases and ash
- Termites and cows → methane
- Electrical discharges in the atmosphere → nitrogen oxides

(iii) Burning fossil fuels (coal, petroleum, natural gas) in vehicles, factories or power plants → carbon monoxide, nitrogen oxides and sulphur oxides. 1

Source: Ch 11, sections 11.2 (p152) and 11.3 (p154). SLO: state the major sources of air pollutants.

3. Formation of pollutants, balanced with state symbols. 4

(i) $2\text{C}_{(\text{s})} + \text{O}_{2(\text{g})} \rightarrow 2\text{CO}_{(\text{g})}$ (limited oxygen) 1

(ii) $\text{S}_{(\text{s})} + \text{O}_{2(\text{g})} \rightarrow \text{SO}_{2(\text{g})}$ 1

(iii) $\text{N}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightarrow 2\text{NO}_{(\text{g})}$ 1

(iv) $2\text{NO}_{(\text{g})} + \text{O}_{2(\text{g})} \rightarrow 2\text{NO}_{2(\text{g})}$ 1

Marking note: each mark is $\frac{1}{2}$ for the balanced equation and $\frac{1}{2}$ for state symbols. **Her book prints (i) unbalanced** as " $\text{C} + \text{O}_2$ (limited) → CO ". The question asks for a **balanced** equation, so the 2 in front of C and CO is required; copying the book's version loses the balancing $\frac{1}{2}$. Source: Ch 11, section 11.3.2, book p154.

4. Carbon monoxide. 4

(i) By **incomplete combustion**, when there is **not enough oxygen** for the fuel to burn completely. 1

(ii) When inhaled, carbon monoxide **combines with haemoglobin** to form **carboxyhaemoglobin** 1, which cannot carry oxygen. The blood cells cannot absorb oxygen, so the person loses consciousness, suffocates and can die. 1

(iii) Carbon monoxide is a **colourless and odourless** gas. 1

Source: Ch 11, section 11.2.4, book p153. SLO: carbon monoxide from incomplete combustion; carbon monoxide as a toxic gas.

5. Global warming. 4

(i) Sunlight reaches the Earth and is absorbed by the surface, which then gives off energy as **infrared radiation**. Greenhouse gases such as carbon dioxide and methane **absorb and re-radiate** some of this infrared back, acting like a **blanket**, so less heat is lost to space 1. Human activities (burning fossil fuels, deforestation, livestock) increase these gases, so more heat is trapped and the Earth's temperature rises: this is global warming. 1

(ii) Any **two**, 1 mark each:

- The temperature of the Earth will gradually increase.
- The climate may change, affecting rainfall: flooding in some regions and drought in others.
- Polar ice may melt and cause a significant rise in sea levels.
- The atmosphere becomes hotter.

Source: Ch 11, section 11.4 and Fig. 11.3, book p155–156. SLO: explain how CO_2 and methane cause global warming.

6. Ground-level ozone. 4

(i) It is **not released directly** from any source; it is **produced by reactions between other pollutants** in sunlight. 1

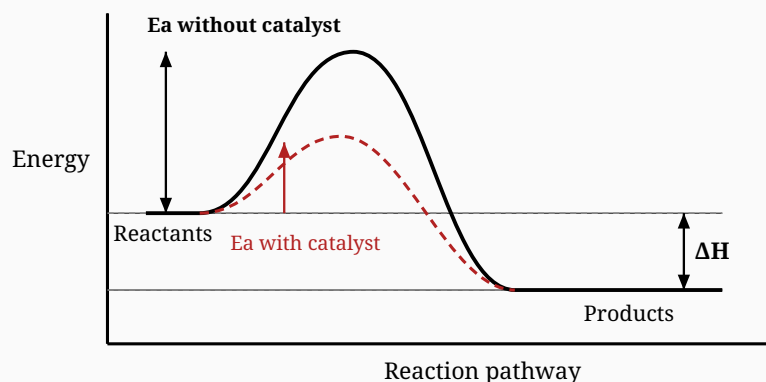
(ii) In sunlight, nitrogen dioxide breaks up: $\text{NO}_2 \rightarrow \text{NO} + \text{O}$ 1

The atomic oxygen then joins an oxygen molecule: $\text{O} + \text{O}_2 \rightarrow \text{O}_3$ 1

(iii) **Nitrogen oxides and volatile organic compounds (VOCs).** 1 ($\frac{1}{2}$ each)

Source: Ch 11, section 11.2.6, book p153–154. SLO: ground-level ozone from reactions of oxides of nitrogen and VOCs in the presence of light. Note (ii): the **atomic** oxygen O reacts with the **molecule** O_2 ; this is the one place in the chapter where a single O is correct.

7. [REVISION, Ch 8] Energy profile with and without a catalyst. 4



Solid black curve = without catalyst. **Red dashed curve** = with catalyst. Products sit **lower** than reactants (exothermic), so ΔH is negative.

(i) Marks: both axes labelled $\frac{1}{2}$; reactants and products labelled, products lower $\frac{1}{2}$; both humps, each with its E_a drawn **from the reactant level up to the top of that hump** 1; ΔH labelled between the reactant and product levels 1.

(ii) “A catalyst **lowers the activation energy** but **does not change ΔH .**” 1

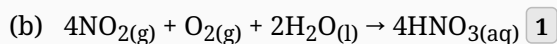
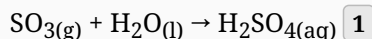
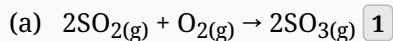
Source: Ch 8, Fig. 8.1 (p130) and Fig. 8.3 with section 8.6 (p131): “The catalyst does not affect the overall thermodynamics or enthalpy of the reaction.” Re-test of exam-23 Q9, where ΔH and this sentence were both missing.

SECTION C: ANSWER KEY (2 × 8 = 16 MARKS)

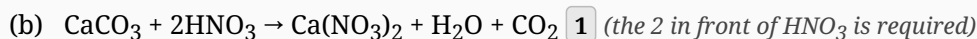
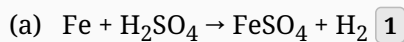
8. Acid rain. 8

(i) Normal rainwater has a **pH of 5.6** 1. Rain with a **pH less than 5.6** is called acid rain. 1

(ii) Each equation 1 mark (½ balanced, ½ state symbols):



(iii)



(iv) Any **two**, ½ each: acid rain **kills fish**; it **destroys trees**; lakes and rivers may become **too acidic for living things to survive**. 1

Source: Ch 11, section 11.5 with Figs 11.4–11.6, book p156–157, which prints all five equations. SLOs: role of sulphur in the formation of acid rain and its impact; role of NO and NO₂ in acid rain. None of these parts appeared on exam-23.

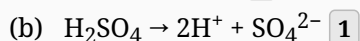
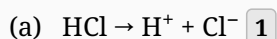
9. [REVISION, Ch 10] Bronsted-Lowry acids and bases, and naming. 8

(i) 1 mark for each correct identification **with** its reason:

Reaction	Bronsted-Lowry acid	Bronsted-Lowry base
(a) $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$	CH₃COOH : donates a proton and becomes CH ₃ COO [−]	H₂O : accepts the proton and becomes H ₃ O ⁺
(b) $\text{H}_2\text{O} + \text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$	H₂O : donates a proton and becomes OH [−]	NH₃ : accepts the proton and becomes NH ₄ ⁺

Water is the base in (a) and the acid in (b): that is why it is called amphoteric. The acid is always on the **left** of the arrow; H₃O⁺ and NH₄⁺ are products.

(ii) Strong acids, so single arrows:



Written in reverse (ions forming the acid) earns 0. Missing charges lose ½.

(iii) ½ each: Ca(OH)₂ **calcium hydroxide**; Mg(OH)₂ **magnesium hydroxide**; Na₂O **sodium oxide**; MgO **magnesium oxide**. 2

Source: Ch 10, Example 10.1 (p143), Concept Assessment 10.1 (p144), sections 10.1.1 and 10.2.1 (p141, p144), section 10.5 (p146). Re-tests of exam-23 MCQ 1 (water as acid), Q2(ii) (product named as the acid), Q3 (ionization reversed) and Q8 (formulas given where names were asked).

Mark distribution check

Section	Chapter 11 (new)	Revision	Total
A: MCQs	7 (Q1–Q7)	3 (Q8–Q10, Ch 10)	10
B: Short	20 (Q2–Q6)	4 (Q7, Ch 8)	24
C: Long	8 (Q8)	8 (Q9, Ch 10)	16
Total	35 (70%)	15 (30%)	50

Answer letter balance for Section A: a = 3 (Q2, Q6, Q10), b = 3 (Q1, Q5, Q8), c = 2 (Q3, Q7), d = 2 (Q4, Q9).